CPSC 100

Computational Thinking

Data Representation

Instructor: Firas Moosvi
Department of Computer Science
University of British Columbia

Agenda

- Course Admin
- Learning Goals
- Intro to Data Representation

Course Admin

Course Admin

- Test 2 released back to you!

- Test 3 released back to you!

- I changed the labs so that you can still view it after you submit them - sorry about that, this was always the intention!

- Lab grading underway…

- Go to your labs this week! Project groups will be made by Friday!

Learning Goals

After this **today's lecture**, you should be able to:

- Understand the usage of data before and after computers.
- Recognize **hexadecimal numbers** and their role in data representation.
- Count in different **standard number bases** (i.e. 2, 10, 16).
- **Translate** numbers between binary, hexadecimal, and decimal without a calculate

After watching the **take home-video**, you should be able to:

- Recognize the difference between binary and ternary numbers
- Translate numbers between binary, hexadecimal, decimal and **ternary**

Q: What will the following expression report?

iClicker

A. True

B. False

Q: What will the following expression report?

iClicker

A. True

B. False

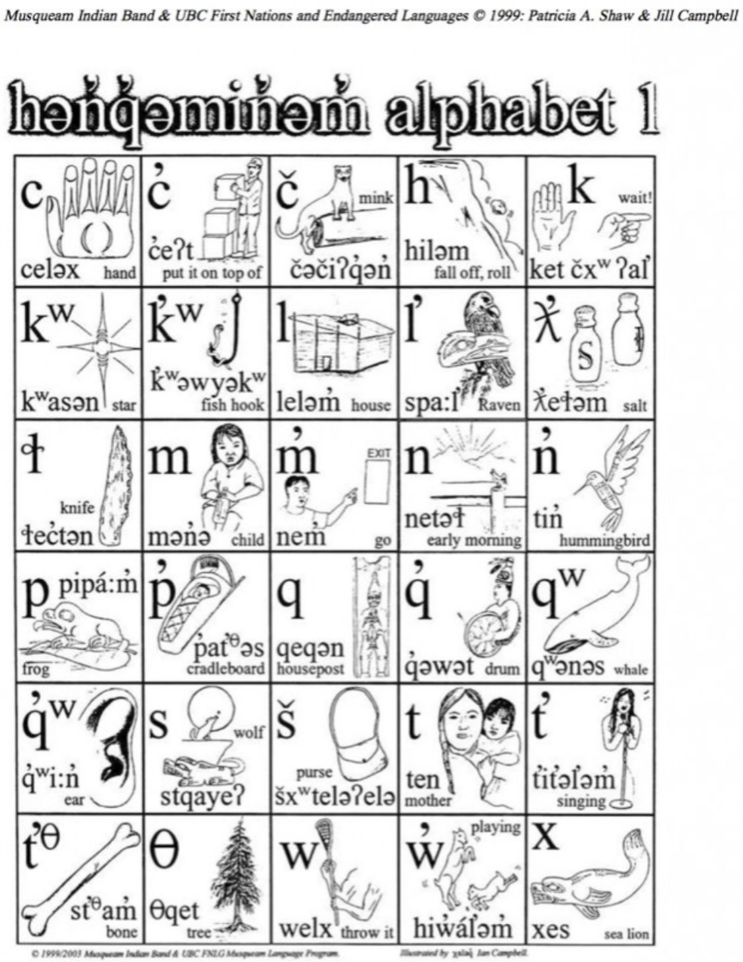
Data Representation

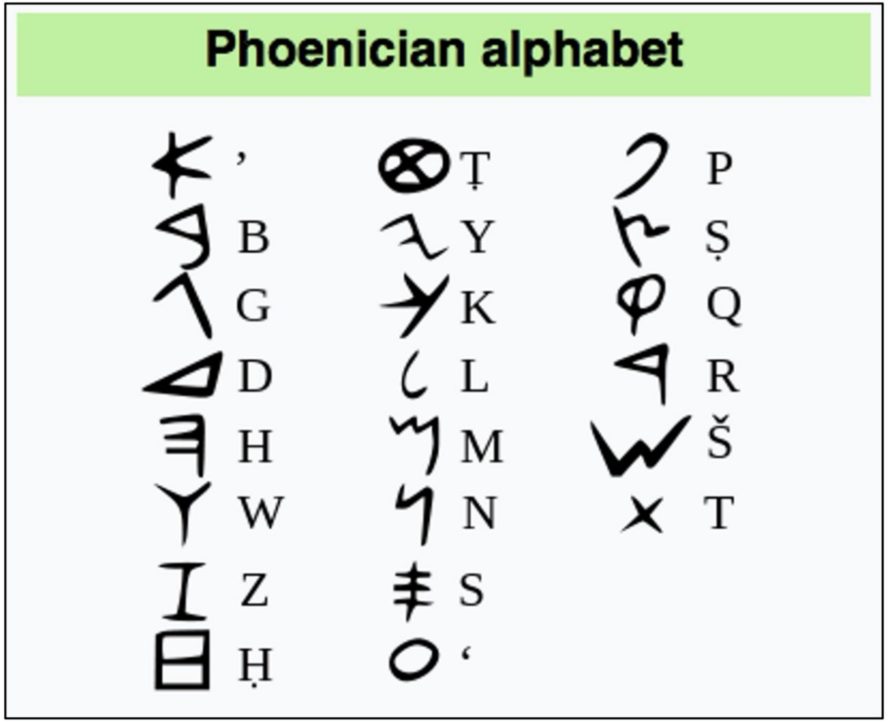
Data Representation Before Computers

Digital Data Before Computers

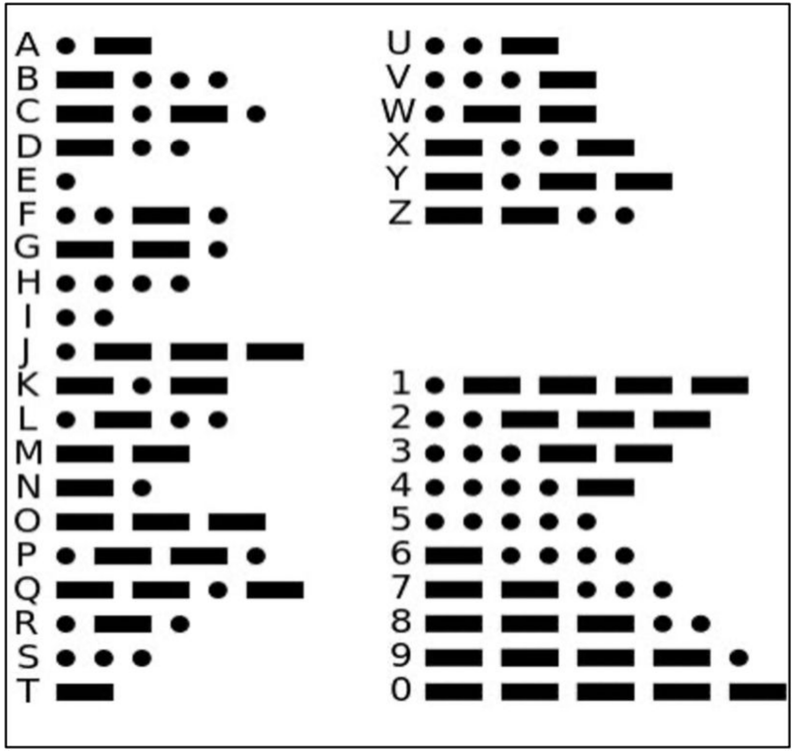
Creative computational thinkers have invented digital data representation systems for millennia, adapting to many different technologies:

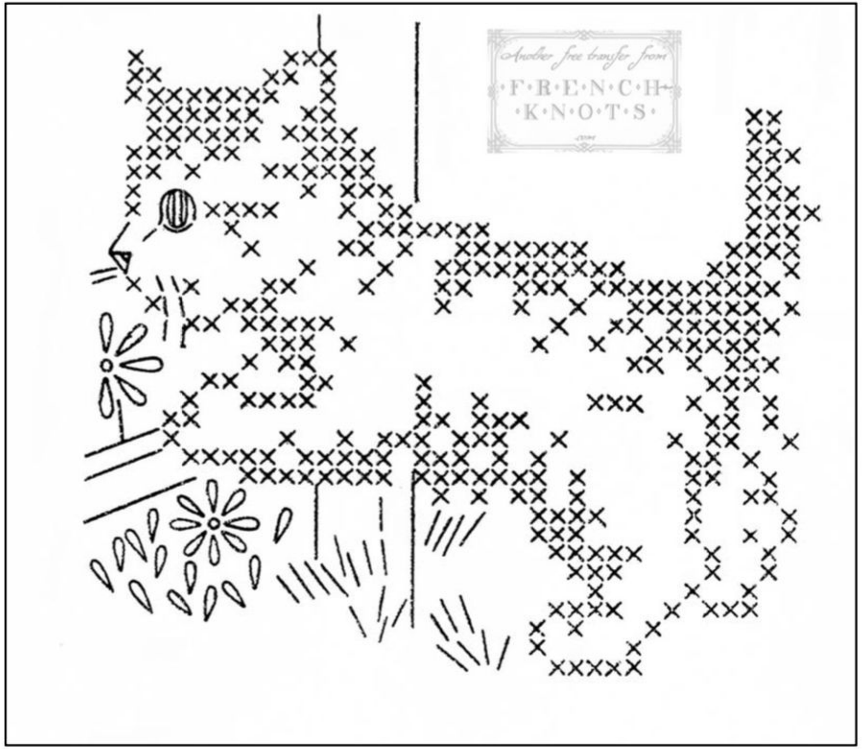
Image source: Wikipedia





Source: Musqueam Teaching and Learning





Morse Code

Cross Stitch

Image source: Wikipedia

Participation Question

How do you count from 1 to 10 on your hands?

Data Representation in nature!

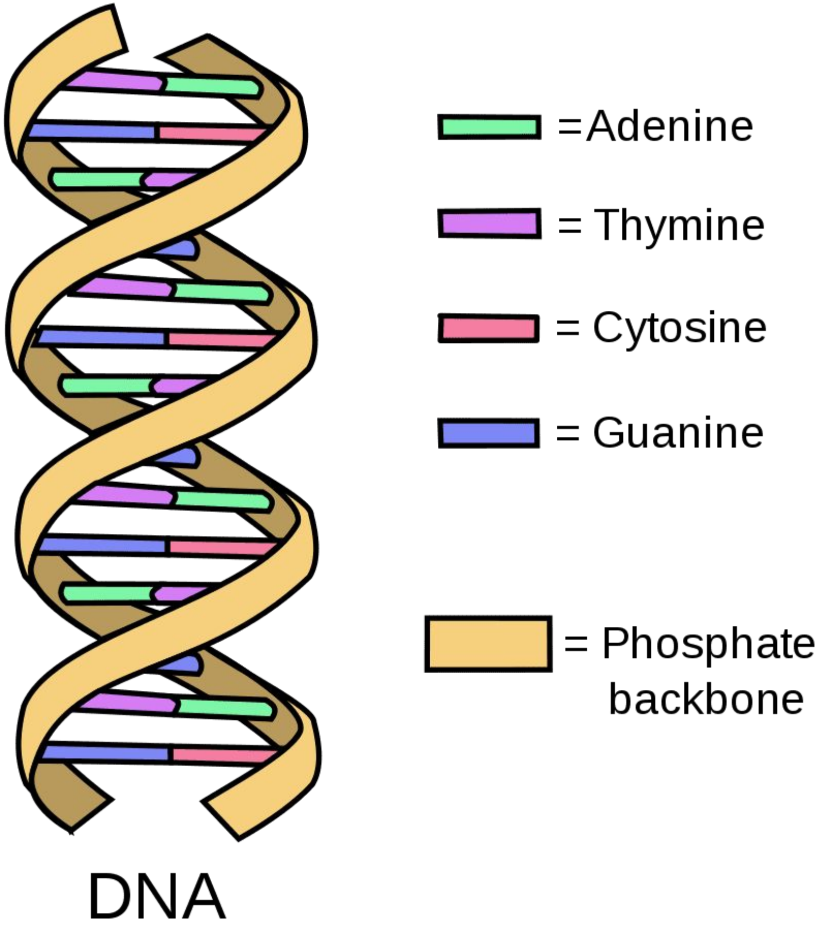


Image source: ashg.org

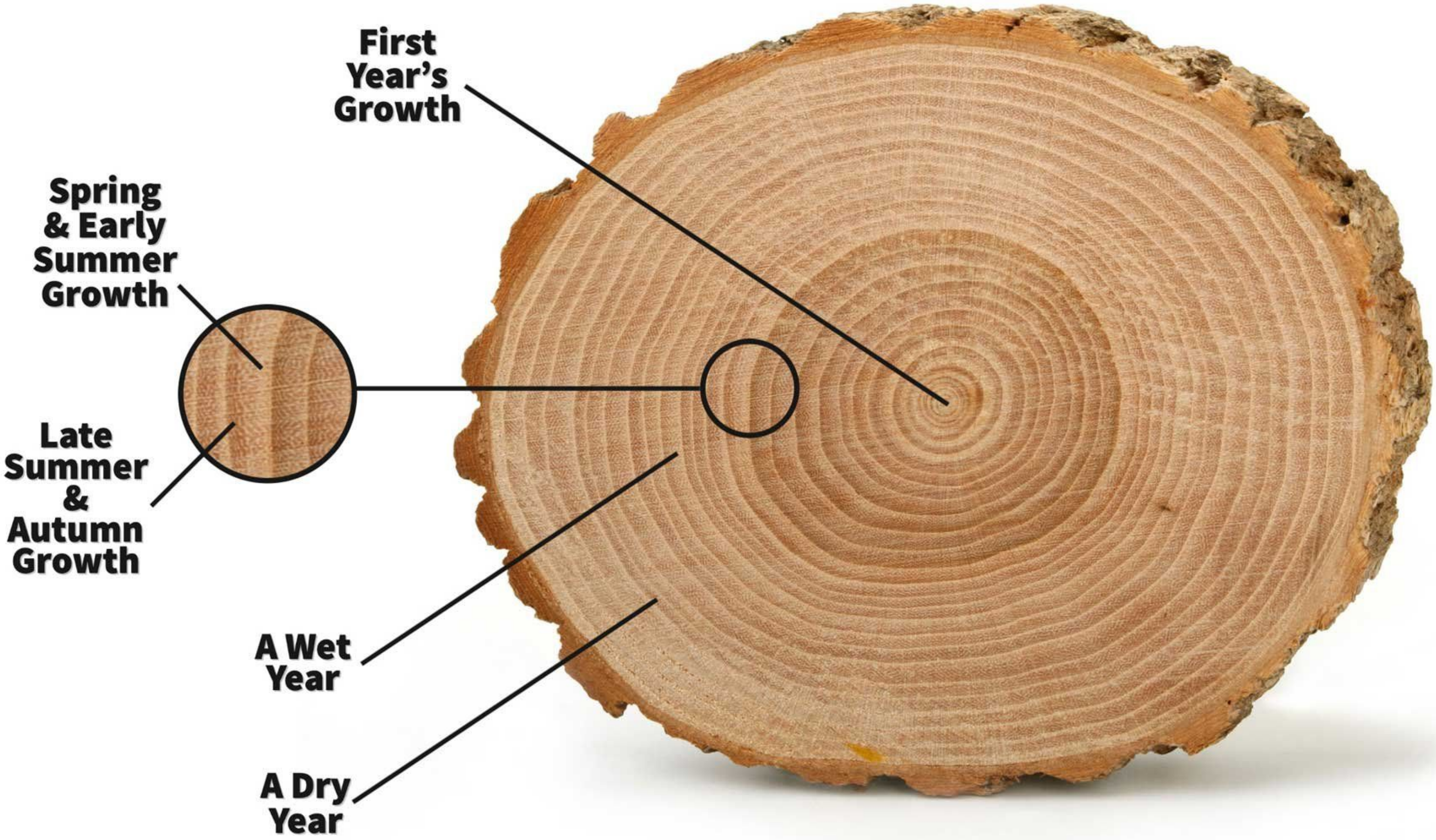


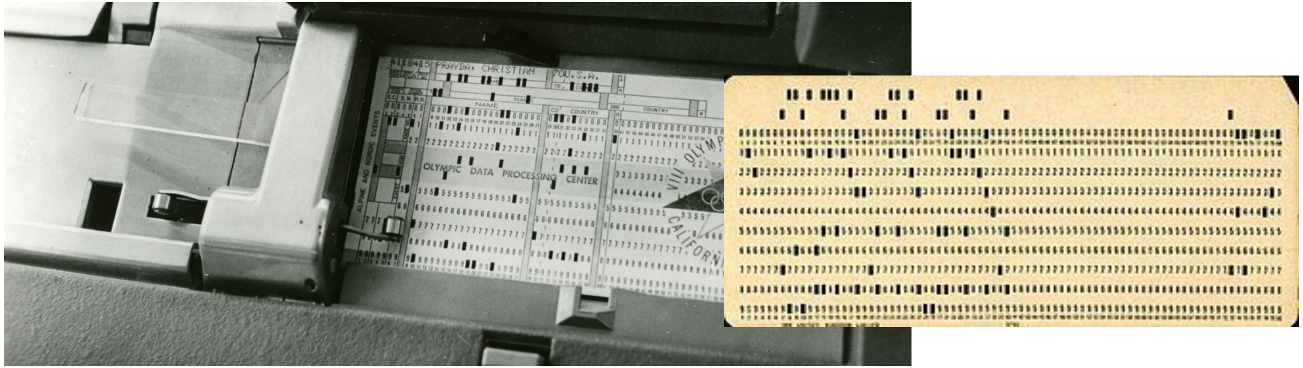
Image source: silvotherapy.co.uk

Data Representation with computers

Digital Data with Computers

- Computing technologies represent numbers, text, images, video, and sound using just two symbols: **0's and 1's**

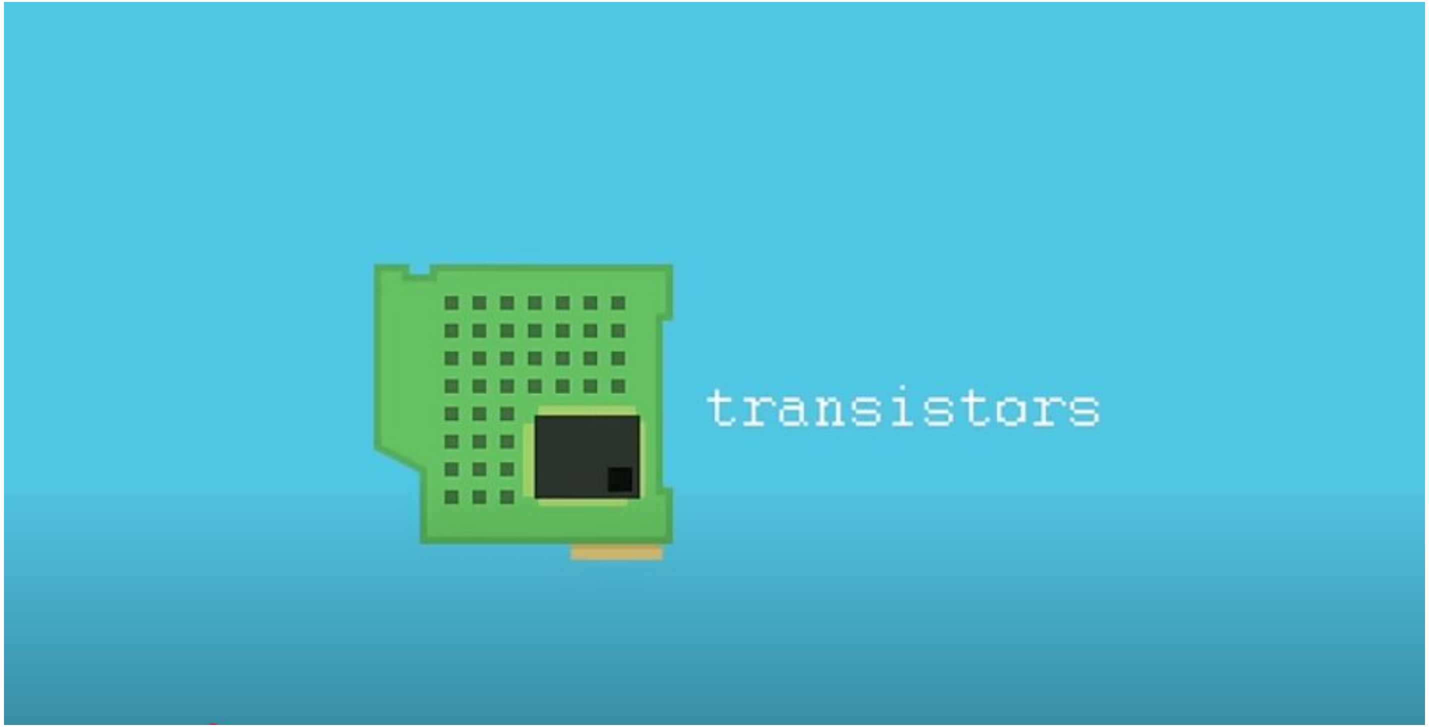
- 0's and 1's are embodied as "high" (on) or "low" (off) signals on various electronic and optical storage media
(e.g., punch cards, vacuum tubes, transistors, DVDs, etc.)



high	low
on	off
true	false
yes	no
+	-
1	0

Image source: Wikipedia

Why is data represented in binary for computers?



Numbers in a Digital Era

- [Base-10] Decimal Numbers
  - E.g., 1, 2, 3, etc.
- [Base-2] Binary Numbers
  - E.g., 11001, 1001, 1111, etc.
- [Base-3] Ternary Numbers
  - E.g., 12_3, 110_3, 20_3, etc.
- [Base-16] Hexadecimal Numbers
  - E.g., FF5733, AB4151, etc.

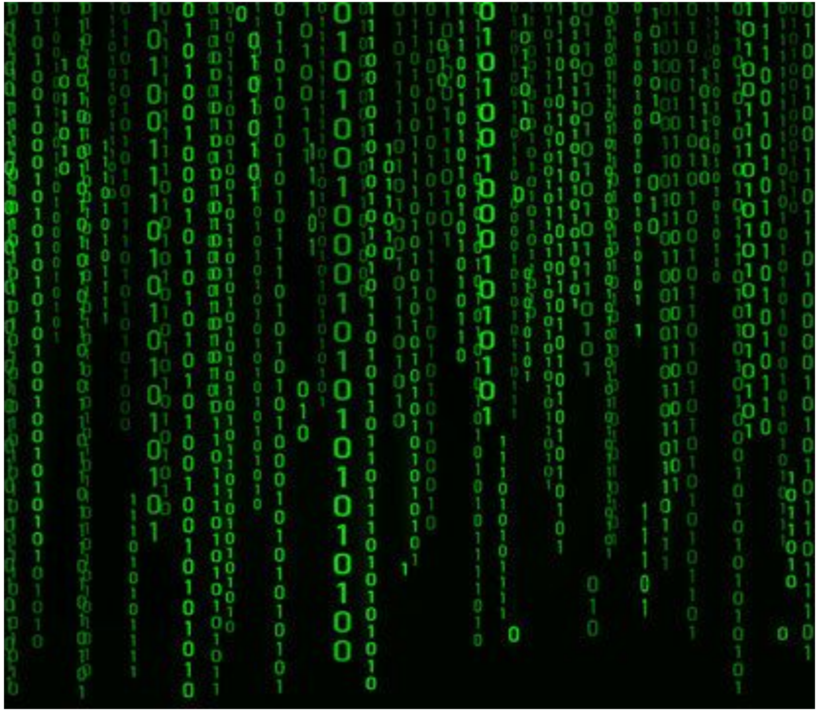


Image source: Adobe

Decimal Numbers

Decimal Numbers

Decimal numbers are base 10.

("decem" is Latin for 10).

In decimal, every column is worth 10 times as much as the previous one

2 4 8 ← Each digit can be any one of these 10 values: {0, 1, 2, 3, 4, 5, 6, 7, 8, 9}

- 8 → 1s place (10^0)
- 4 → 10s place (10^1)
- 2 → 100s place 100 (10^2)

Counting in Decimal

When counting in decimal:

- Add one to the right most digit
- If that digit has the highest value (9), then add one to the digit in the next column to the left and reset the column you're on to zero.
- If you run out of digits in the column to the left, you repeat the process

- 0
- 1
- 2
- 3
- 4
- 5
- 6
- 7
- 8
- 9

...

99

100

...

10999

11000

Counting in Decimal Example 1

$$\underline{\ 1\ }\quad \underline{\ 0\ }\quad \underline{\ 0\ }\quad +1 \rightarrow \quad \underline{\ 1\ }\quad \underline{\ 0\ }\quad \underline{\ 1\ }$$

Add 1 to the
rightmost digit

Counting in Decimal Example 2

1 0 9 +1 → 1 1 0

Add 1 to the rightmost digit. 9 is the highest representation for a number, so we have to change 9 to 0 and increment the number to its left by 1

27

Binary Numbers

Bits

- A **bit** is short for a **binary digit**.
  - It is one 0 or 1. E.g., one "high" segment on a DVD or an "on" transistor.
- When a computer sees 1000000 (e.g. one "on" part followed by six "off" parts), it reads this as **binary** and *processes* this as **decimal 64**.
  - 64 and 1000000 are two representations of the same number
- When reading out binary numbers, **we say the digits in turn**
  - For “010” we say "zero one zero" or just "one zero“ (We don't say "ten").
  - Sometimes we write this as **0b010** to make it clear it is a **binary** number.
  - You may also see a subscript 2 after the number (e.g., 010_2).

Binary Numbers

Binary numbers are base 2.

("binarius" is Latin for having two parts).

In binary, every column is worth 2 times as much as the previous one

1 1 0 0 1 ← Binary

1s place (2^0)

2s place (2^1)

4s place (2^2)

8s place (2^3)

16s place (2^4)

Convert Binary to Decimal

0b 1 1 0 0 1

1s place multiply by 1	(2^0):	1
2s place multiply by 2	(2^1):	0
4s place multiply by 4	(2^2):	0
8s place multiply by 8	(2^3):	8
16s place multiply by 16	(2^4):	+ 16
Total in decimal (add them up):		25

Algorithm to Convert Decimal to Binary

- Start with a decimal number *n* (0 to 255)
  - Since 255 is the maximum, we need 8 bits.
- Check the highest power of 2 that fits into *n*:
  - If n < 128, set the 1st (leftmost) bit to 0; otherwise, set it to 1 and subtract 128 from *n* (*n=n-128*)
  - If n < 64, set the 2nd bit to 0; otherwise, set it to 1 and subtract 64 from *n*. (*n=n-64*)
  - If n < 32, set the 3rd bit to 0; otherwise, set it to 1 and subtract 32 from *n*. (*n=n-32*)
  - Continue this process for 16, 8, 4, 2, and finally 1.
- By the end, *n* will be either 0 or 1, which is the last (8th) bit.

Algorithm to Convert 217 to Binary

- n = 217
- Check the highest power of 2 that fits into *n*:
  - ❌ **217** < 128 (2^7), 1st Bit = 1
    - *n = 217 - 128* ⇒ 89
  - ❌ **89** < 64 (2^6), 2nd bit = 1
    - *n = 89 - 64 ⇒ 25*
  - ✅ **25** < 32 (2^5), 3rd bit = 0
    - *No subtraction required*
  - ❌ **25** < 16 (2^4), 4nd bit = 1
    - *n = 25 - 16* ⇒ 9
  - ❌ **9** < 8 (2^3), 5th Bit = 1
    - *n = 9 - 8* ⇒ 1
  - ✅ **1** < 4 (2^2), 6th bit = 0
    - *No subtraction required*
  - ✅ **1** < 2 (2^1), 7th bit = 0
    - *No subtraction required*
  - **Final bit = n = 1 (2^0)→ 8th bit =** 1
- **Final Answer: 11011001**

Binary Numbers

- Binary numbers only use two digits: 0 and 1, and their place values are based on powers of 2.

Number 9 in Binary:

256	128	64	32	16	8	4	2	1
0	**0**	**0**	**0**	**0**	**1**	**0**	**0**	**1**

9 = 000001001

Counting in Binary

When counting in binary:

- Add one to the right most digit
- If that digit has the highest value (1), then add one to the digit in the next column to the left and reset the column you're on to zero.
- If you run out of digits in the column to the left, you repeat the process

- 0
- 1
- 10
- 11
- 100
- 101
- …
- 101101
- 101110
- 110001

Counting in Binary Example 1

1 0 0 +1 → 1 0 1

Add 1 to the
rightmost digit

Clicker: Addition in Binary

iClicker

1 0 1 +1 → ? ? ?

What number should go on the right side?
(note: both #s are in binary)

A) 100
B) 101
C) 110
D) 111

Clicker: Addition in Binary

iClicker

1 0 1 +1 → ? ? ?

What number should go on the right side?
(note: both #s are in binary)

A) 100
B) 101
C) 110
D) 111

Q: The 8-bit binary representation of 57 is 00111001. What is the 8-bit binary representation of 58?

iClicker

A. 01011110

B. 00111111

C. 00111010

D. 0011100010001

Q: The 8-bit binary representation of 57 is 00111001. What is the 8-bit binary representation of 58?

iClicker

A. 01011110

B. 00111111

C. 00111010

D. 0011100010001

Wrap up

Parsa - Week 6A

CPSC 100

Computational Thinking

Intro to Data Representation

Instructor: Parsa Rajabi
Department of Computer Science
University of British Columbia

Agenda

- Learning Goals
- Course Admin
- Intro to Data Representation
  - Activity
  - Counting in Binary

Learning Goals

After this today's lecture, you should be able to:

- Recognize binary’s role in data representation
- Understand that binary (base-2) is the fundamental numbering system used in computing.
  - Explain why computers use binary
- Convert given decimal numbers (e.g., 13) into binary.

Course Admin

Course Admin

- **Project Milestone 1**
  - Due on Wednesday, Feb 12 at 11:59pm
- **Midterm**
  - Review exam details here
  - You're allowed a cheat sheet: **1 side of an A4 sized paper**
  - 📅 February 14 💘
  - ⏰ 3:00-3:50 pm
  - !!⚠️ CHBE 101 [**Ch**emical and **B**iological **E**ngineering Building] ⚠️!!
  - Practice Problems available here
  - Past Exams posted on Canvas

Data Representation

Take-Home Practice

Convert these numbers to binary

- 16
- 27
- 30
- 58
- 98
- 78
- 100
- 9
- 34

What was your main takeaway from today's session?

Do not edit
How to change the design

Presenting with animations, GIFs or speaker notes? Enable our Chrome extension

slido

Wrap up

Wrap Up

- **Project Milestone 1**
  - Due on Wednesday, Feb 12 at 11:59pm
- **Midterm**
  - Review exam details here
  - You're allowed a cheat sheet: **1 side of an A4 sized paper**
  - 📅 February 14 💘
  - ⏰ 3:00-3:50 pm
  - ‼️⚠️ CHBE 101 [**Ch**emical and **B**iological **E**ngineering Building] ⚠️‼️
  - Practice Problems available here
  - Past Exams posted on Canvas

Parsa - Week 8A

Q: What is the binary number for 9?

iClicker

A. 10001
B. 01010
C. 01011
D. 01000
E. 01001

Q: What is the binary number for 9?

iClicker

A. 10001
B. 01010
C. 01011
D. 01000
E. 01001

CPSC 100

Computational Thinking

Data Representation *Continued*

Instructor: Parsa Rajabi
Department of Computer Science
University of British Columbia

Agenda

- Course Admin
- Learning Goals
- Data Representation *Continued*
  - Binary Numbers [*Review*]
  - Decimal Numbers
  - Hexadecimal

Course Admin

Course Admin

- Parsa (*Instructor*) away for conference [SIGCSE 2025]
  - **Wednesday, Feb 26 → Monday, Mar 3 (inclusive)**
  - **Wednesday, Feb 26** - Kate (TA)
  - **Friday, Feb 28** - Kate/Parsa (TA)
  - **Monday, Mar 3** - Kevin Lindstrom (UBC Librarian)
- **Lab 5** - Number base system in Snap!
  - Due Friday, Feb 28 (extra day provided due midterm week)
- **PC Quiz 4**
  - Released Monday, March 3
  - Due Sunday, March 9

Learning Goals

Kibbles and Bits

Kibbles and bits

- A **bit** is short for a **binary digit**.
  - It is one 0 or 1. E.g., one "high" segment on a DVD or an "on" transistor.
- When a computer sees 1000000 (e.g. one "on" part on a DVD followed by six "off" parts on a DVD), it reads this as **binary** and *processes* this as **decimal 64**.
  - 64 and 1000000 are two representations of the same number
- When reading out binary numbers, **we say the digits in turn**
  - For “010” we say "zero one zero" or just "one zero“ (We don't say "ten").
  - Sometimes we write this as **0b010** to make it clear it is a **binary** number.
  - You may also see a subscript 2 after the number (e.g., 010_2).

Convert Binary to Decimal

Recall

0b 1 1 0 0 1

1s place multiply by 1	(2^0):	1
2s place multiply by 2	(2^1):	0
4s place multiply by 4	(2^2):	0
8s place multiply by 8	(2^3):	8
16s place multiply by 16	(2^4):	+ 16
Total in decimal (add them up):		25

Algorithm to Convert Decimal to Binary

- Start with a decimal number *n* (0 to 255)
  - Since 255 is the maximum, we need 8 bits.
- Check the highest power of 2 that fits into *n*:
  - If n < 128, set the 1st (leftmost) bit to 0; otherwise, set it to 1 and subtract 128 from *n* (*n=n-128*)
  - If n < 64, set the 2nd bit to 0; otherwise, set it to 1 and subtract 64 from *n*. (*n=n-64*)
  - If n < 32, set the 3rd bit to 0; otherwise, set it to 1 and subtract 32 from *n*. (*n=n-32*)
  - Continue this process for 16, 8, 4, 2, and finally 1.
- By the end, *n* will be either 0 or 1, which is the last (8th) bit.

Algorithm to Convert 217 to Binary

- n = 217
- Check the highest power of 2 that fits into *n*:
  - ❌ **217** < 128 (2^7), 1st Bit = 1
    - *n = 217 - 128* ⇒ 89
  - ❌ **89** < 64 (2^6), 2nd bit = 1
    - *n = 89 - 64 ⇒ 25*
  - ✅ **25** < 32 (2^5), 3rd bit = 0
    - *No subtraction required*
  - ❌ **25** < 16 (2^4), 4nd bit = 1
    - *n = 25 - 16* ⇒ 9
  - ❌ **9** < 8 (2^3), 5th Bit = 1
    - *n = 9 - 8* ⇒ 1
  - ✅ **1** < 4 (2^2), 6th bit = 0
    - *No subtraction required*
  - ✅ **1** < 2 (2^1), 7th bit = 0
    - *No subtraction required*
  - **Final bit = n = 1 (2^0)⟶ 8th bit =** 1
- **Final Answer: 11011001**

69

Take Home Activity

Convert 365_{10} (Decimal) to Binary

Hexadecimal Numbers

Hexadecimal Numbers

- Computers don't understand Decimal (base 10)
- Humans have trouble with Binary (base 2)
  - 11111111100110001110001010 10
  - Writing so many 0's and 1's is tedious and error prone
- **Hex digits**, short for **hex**adecimal (base 16)
  - Compromise between humans and computers
  - **Easy to convert between Hex and Binary**

Hexadecimal Numbers

- The digits of the hexadecimal numbering system are
  - 0, 1, 2, 3, 4, 5, 6, 7, 8, 9,   A, B, C, D, E, F
- Because there are 16 digits, they can be represented perfectly by the 16 symbols of 4-bit sequences:
  - The bit sequence **0000** is **hex 0** Bit sequence **0001** is **hex 1**
  - Bit sequence 1111, is hex F
  - Sometimes we use the representation **0x[number]** to show that a number is in **hex**.

MetaCog Activity - Fill in the Gaps
(Take Home)

Decimal	Binary	Hex
00	000	0
01	001	1
02	010	
03		
04	100	
05		
06		6
07		

Decimal	Binary	Hex
08		
09	1001	
10		A
11	1011	
12		
13	1101	
14		
15		F

(leading 0's, shown in gray, are useful for some conversions)

Hexadecimal Numbers to Binary

- Because each hex digit corresponds to a 4-binary sequence, it's easy to translate between hex and binary
  - Hex → binary: replace each hexadecimal digit by the four corresponding binary digits in the conversion table

Hex:	F	A	B	4
Binary:	1111	1010	1011	0100

Binary to Hexadecimal

- Binary → hex:
  - Put extra 0's at the left of the binary number as necessary so that the total number of digits is a multiple of 4
  - then reverse the hex → binary conversion process

Binary:	0010	1011	1010	1101
Hex:	2	B	A	D

MetaCog Activity - Fill in the Gaps
(Take Home)

Decimal	Binary	Hex
00	000	0
01	001	1
02	010	2
03	011	3
04	100	4
05	101	5
06	110	6
07	111	7

Decimal	Binary	Hex
08	1000	8
09	1001	9
10	1010	A
11	1011	B
12	1100	C
13	1101	D
14	1110	E
15	1111	F

(leading 0's, shown in gray, are useful for some conversions)

Q: Translate 0xAFF1 to binary

iClicker

A. 1111 1101 1101 0001
B. 1010 1101 1101 0001
C. 0001 1111 1111 1010
D. 1111 1010 1111 0001
E. 1010 1111 1111 0001

Q: Translate 0xAFF1 to binary

iClicker

A. 1111 1101 1101 0001
B. 1010 1101 1101 0001
C. 0001 1111 1111 1010
D. 1111 1010 1111 0001
E. 1010 1111 1111 0001

Take Home Activity

Convert 1100 1010 1111 1110 to hexadecimal

Convert 0xA19C to decimal

Convert 48 to hexadecimal

Take-Home Video (watch before lab)

https://youtu.be/CBYhwcn4WSI?si=fITtLe_y5ZyqDQTA&t=57

What was your main takeaway from today's session?

Wrap up

Wrap Up

- Parsa (*Instructor*) away for conference [SIGCSE 2025]
  - **Wednesday, Feb 26 → Monday, Mar 3 (inclusive)**
  - **Wednesday, Feb 26** - Kate (TA)
  - **Friday, Feb 28** - Kate/Parsa (TA)
  - **Monday, Mar 3** - Kevin (UBC Librarian)
- **Lab 5** - Number base system in Snap!
  - Due Friday, Feb 28 (extra day provided due midterm week)
- **PC Quiz 4**
  - Released Monday, March 3
  - Due Sunday, March 9

Parsa - Week 9B

What was your takeaway from the past few lectures? (Kate/Parsa/Kevin)?

CPSC 100

Computational Thinking

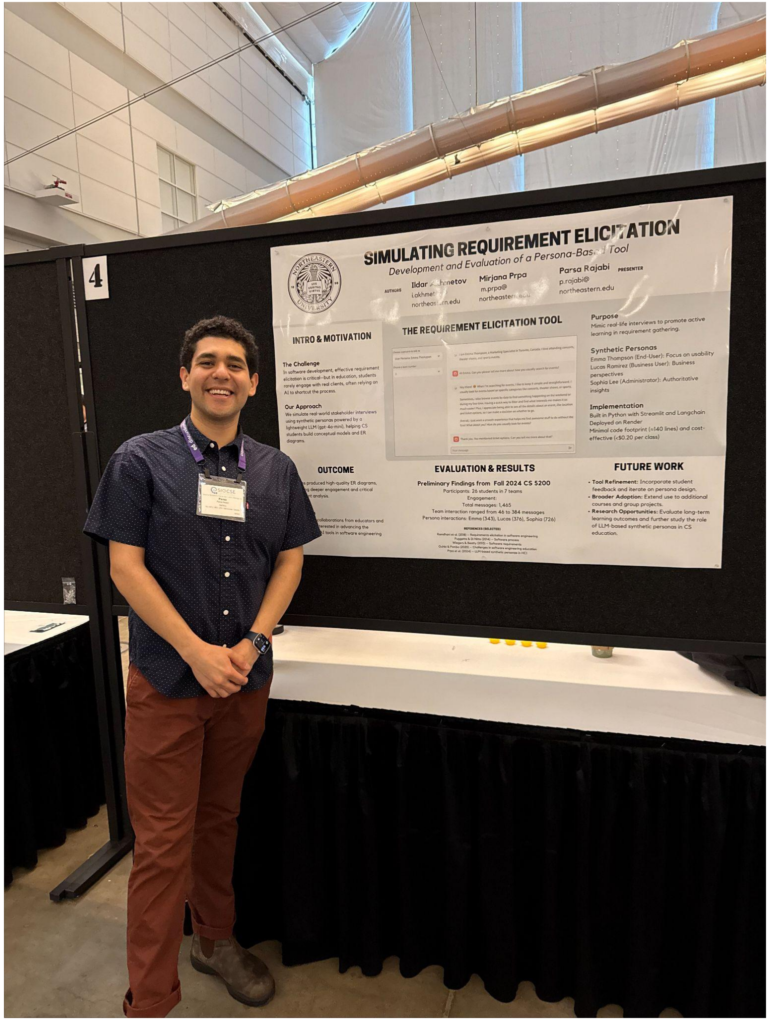
Data Representation *Continued*

Instructor: Parsa Rajabi
Department of Computer Science
University of British Columbia

Agenda

- Conference Takeaways
- Course Admin
- Learning Goals
- Data Representation *Continued*
  - ASCII
  - Unicode

SIGCSE Conference


4
SIMULATING REQUIREMENT ELICITATION
AUTHORS
Mirjana Prpa
m.prpa@
Parsa Rajabi
PRESENTER
p.rajabi@
northeastern.edu
NORTHEASTERN UNIVERSITY
INTRO & MOTIVATION
The Challenge
In software development, effective requirement elicitation is critical—but in education, students rarely engage with real clients, often relying on AI to shortcut the process.
Our Approach
We simulate real-world stakeholder interviews using synthetic personas powered by a lightweight LLM (gpt-4o-mini), helping CS students build conceptual models and ER diagrams.
THE REQUIREMENT ELICITATION TOOL
Purpose
Mimic real-life interviews to promote active learning in requirement gathering.
Synthetic Personas
Emma Thompson (End-User): Focus on usability
Lucas Ramirez (Business User): Business perspectives
Sophia Lee (Administrator): Authoritative insights
Implementation
Built in Python with Streamlit and Langchain
Deployed on Render
Minimal code footprint (≈140 lines) and cost-effective (<$0.20 per class)
OUTCOME
EVALUATION & RESULTS
Preliminary Findings from Fall 2024 CS 5200
Participants: 26 students in 7 teams
Engagement:
Total messages: 1,465
Team interaction ranged from 46 to 384 messages
Persona interactions: Emma (343), Lucas (376), Sophia (726)
FUTURE WORK
• Tool Refinement: Incorporate student feedback and iterate on persona design.
• Broader Adoption: Extend use to additional courses and group projects.
• Research Opportunities: Evaluate long-term learning outcomes and further study the role of LLM-based synthetic personas in CS education.

Course
Admin

Course Admin

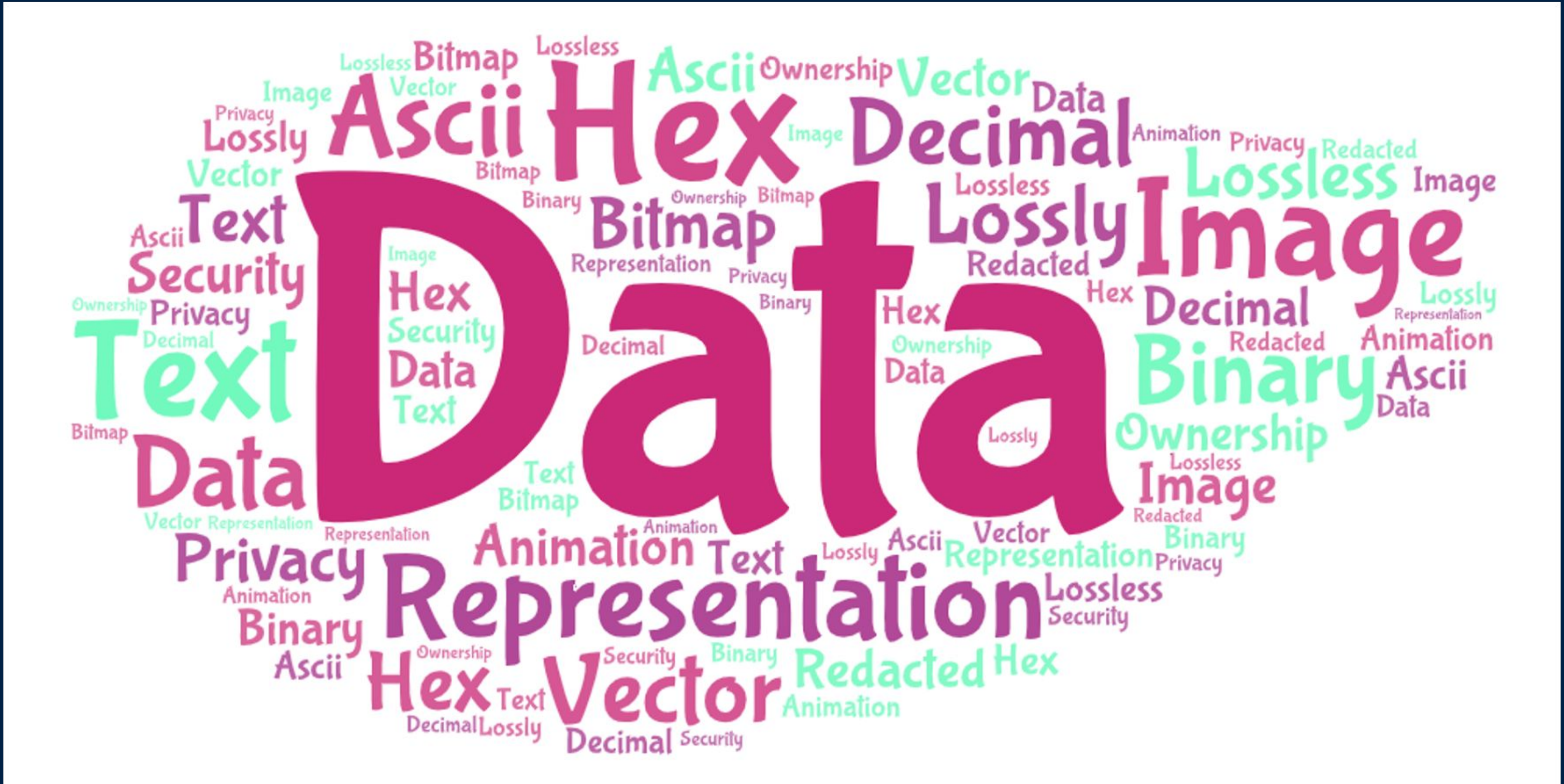
- **Lab 6** - Unicode in Snap!
  - Due Thursday, March 6, 11:59pm
- **PC Quiz 4**
  - Due Sunday, March 9, 11:59pm
- **Midterms Grades**
  - Grading complete; reviewing and finalizing grades
- **Project Milestone 2**
  - Due Wednesday, March 12, 11:59pm
  - **[New!]** *AI Disclosure Form*
- **Final Exam**
  - Tuesday, April 22, 7pm; Location TBA

Learning Goals

Learning Goals

After this **today's lecture**, you should be able to:

- Explain what **ASCII** and **Unicode** are, including their historical context, purpose, and significance in computing.
- **Decode** an **ASCII** representation of a short text document
  - (with a list of ASCII codes provided)
- Articulate **why Unicode was created** and how it solved the problems of earlier encoding systems like ASCII
- **Compare and contrast Unicode with ASCII** in terms of character range, encoding length, and use cases


Lossless Bitmap Lossless Ascii Ownership Vector
Image Vector Data
Privacy Ascii Hex Decimal
Lossly Image Animation Privacy Redacted
Vector Bitmap Lossless Lossless Image
Text Binary Ownership Bitmap
Ascii Bitmap Lossly Image
Security Image Representation Redacted
Hex Privacy
Ownership Privacy Binary Hex Decimal Lossly
Text Security Hex Representation
Decimal Decimal Ownership Redacted Animation
Data Data Binary Ascii
Text Data
Bitmap Lossly Ownership
Data Text Lossless
Bitmap Image
Vector Representation Animation Redacted
Representation Ascii Vector Binary
Privacy Animation Text Lossly Representation Privacy
Representation
Animation Lossless
Binary Security
Ascii Ownership Security Binary Redacted Hex
Hex Text Vector
Animation
Decimal Lossly Decimal Security

Data Representation **in characters!**

How do computer store letters and characters?

ASCII Table

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ASCII: Overview

American **S**tandard **C**ode for **I**nformation **I**nterchange (ASCII)

- Character encoding standard
- Allows computers and electronic devices to represent text
- Developed in 1960s
  - Standardize how computers represent characters

ASCII: Overview

- Punctuation, spaces and other special control characters are also encoded
  - each encoded item is sometimes called a **code point**
- ASCII uses **7 bits** to represent each character, which allows for 128 (2^7) unique characters (from 0 to 127)
  - **Why 7 bits?**
  - An extra "check" bit to detect certain errors that might arise
- *Extended* ASCII uses **8 bits** (or one byte), allowing for characters with accents (Á, ë and others)

ASCII: Overview

ASCII uses **7 bits** to represent each character:

- **Control characters** (0–31): For managing hardware (like line breaks or bell sounds).
- **Printable characters** (32–126): Letters, digits, punctuation, and symbols.
- **Delete character** (127).

ASCII: Control characters (0-31)

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ASCII: Printable characters (32-126)

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

ASCII: Delete character (127)

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[ENG OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Q: Convert Hi! from ASCII to Decimal

iClicker

A. 48, 69, 21

B. 47, 68, 20

C. 71, 104, 34

D. 72, 105, 33

E. None of the above

Q: Convert Hi! from ASCII to Decimal

iClicker

A. 48, 69, 21

B. 47, 68, 20

C. 71, 104, 34

D. 72, 105, 33

E. None of the above

Q: Convert the following Hex to ASCII:

iClicker

A. BADGE
B. *),/-
C. A@CFD
D.)(+.,
E. None of the above

Binary	01000010	01000001	01000100	01000111	01000101
Hex	42	41	44	47	45
ASCII					

Q: Convert the following Hex to ASCII:

iClicker

A. BADGE

B. *),/-

C. A@CFD

D.)(+.,

E. None of the above

Binary	01000010	01000001	01000100	01000111	01000101
Hex	42	41	44	47	45
ASCII					

Can we do this faster?

Convert Text to ASCII

- You can use online hex editors to translate from hex to ASCII
  - https://hexed.it/

Mini-Activity

- Create a text file on your computer
- Write your name in the file as plaintext
- Open a file using *Open file* feature of hexed.it
- What do you notice?

Demo

Application of ASCII

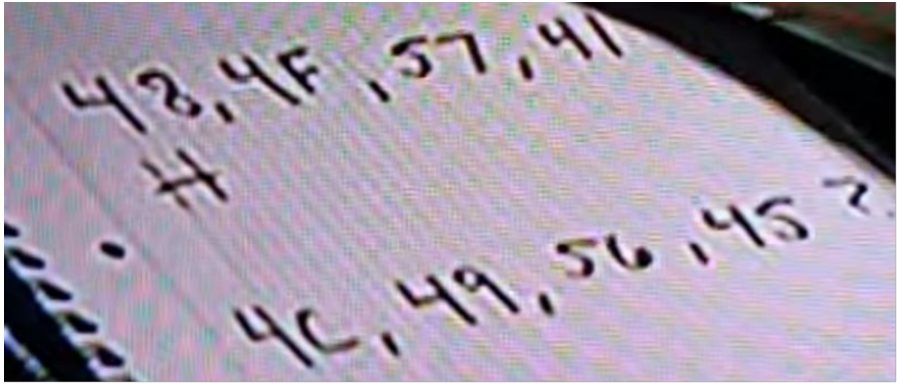
The Martian

The Martian

- “Ridley Scott claimed that one of the most difficult scenes to direct was how to explain to the audience the hexadecimal system Watney uses as a code to communicate with Earth, which Scott admitted was hard for himself to understand.” IMDb

- Youtube Video:
  - The Martian, Hexadecimal Scene

The Martian



Decimal	Hex	Char
64	40	@
65	41	A
66	42	B
67	43	C
68	44	D
69	45	E
70	46	F
71	47	G
72	48	H
73	49	I
74	4A	J
75	4B	K
76	4C	L
77	4D	M
78	4E	N
79	4F	O
80	50	P
81	51	Q
82	52	R
83	53	S
84	54	T
85	55	U
86	56	V
87	57	W
88	58	X
89	59	Y
90	5A	Z

What about other languages?

Arabic (ا-ي)

Chinese (汉字)

Emojis (😊, 🚀)

Unicode

Unicode

Unicode: Overview

Universal character **encoding** standard

- Designed to represent every character from every language in the world, as well as symbols, emojis, and special scripts, using a unified system.

Before unicode…

- Different encoding systems (like ASCII, ISO-8859-1, Shift-JIS)
  - Difficult to mix languages in one document
- Unicode solved this by creating **one global codebook**

How does Unicode Work?

- Unicode assigns a **unique number** called a "**code point**" to every character, regardless of platform, program, or language.
  - Code points are written like: U+0041 (which is 'A').

- Unicode itself is just a standard. To store the characters in files and transmit them over networks, you need **encoding formats**
  - There are different implementations, including UTF-8 and UTF-16 (UTF stands for **U**nicode **T**ransformation **F**ormat)

Encoding Formats

- UTF-8 and UTF-16 are variable length encodings
- They use 1 byte (8 bits) for ASCII, but more for other characters

character	encoding	bits
A	UTF-8	01000001
A	UTF-16	00000000 01000001
A	UTF-32	00000000 00000000 00000000 01000001
あ	UTF-8	11100011 10000001 10000010
あ	UTF-16	00110000 01000010
あ	UTF-32	00000000 00000000 00110000 01000010

Activity

Activity

How does Microsoft Word store its data?

- Open Microsoft Word
- Write your name in the file and save it
- Visit https://hexed.it/
- Open the file using *Open file* feature
- What do you notice?

How does Word store its data?

- Uploading a Word document into the online Hex editor suggests that the document is not in ASCII representation

- Most of the files that comprise a Word document are in **XML** (**Ex**tensible **M**arkup **L**anguage) format; they describe metadata such as the font style and size, document creator, etc.

- The files may also contain information about tracked changes to the document, collaborators, privacy and security settings, and more

Take Home Activity

Convert your name into its ASCII values

Convert your name into its Hex format

Convert your name into its Binary format

Use https://symbl.cc/ to find the unicode for the following characters:

- 🚀
- 🇨🇦
- 🐼
- @
- ê

Project
AI Disclosure Form

Do not edit
How to change the design

What was your main takeaway from today's session?

Presenting with animations, GIFs or speaker notes? Enable our Chrome extension

slido

Wrap up

Wrap Up

- **Lab 6** - Unicode in Snap!
  - Due Thursday, March 6, 11:59pm
- **PC Quiz 4**
  - Due Sunday, March 9, 11:59pm
- **Midterms Grades**
  - Grading complete
  - Reviewing and finalizing grades
- **Project Milestone 2**
  - Due Wednesday, March 12, 11:59pm
  - **[New!]** *AI Disclosure Form*

Parsa - Week 10A

Q: Convert the following hexadecimal sequence to ASCII: 53 54 41 52 53

iClicker

A. STARS

B. 56)45

C. !@#$$

D. HELLO

E. WORLD

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

Q: Convert the following hexadecimal sequence to ASCII: 53 54 41 52 53

iClicker

A. STARS

B. 56)45

C. !@#$$

D. HELLO

E. WORLD

Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char	Decimal	Hex	Char
0	0	[NULL]	32	20	[SPACE]	64	40	@	96	60	`
1	1	[START OF HEADING]	33	21	!	65	41	A	97	61	a
2	2	[START OF TEXT]	34	22	"	66	42	B	98	62	b
3	3	[END OF TEXT]	35	23	#	67	43	C	99	63	c
4	4	[END OF TRANSMISSION]	36	24	$	68	44	D	100	64	d
5	5	[ENQUIRY]	37	25	%	69	45	E	101	65	e
6	6	[ACKNOWLEDGE]	38	26	&	70	46	F	102	66	f
7	7	[BELL]	39	27	'	71	47	G	103	67	g
8	8	[BACKSPACE]	40	28	(	72	48	H	104	68	h
9	9	[HORIZONTAL TAB]	41	29	)	73	49	I	105	69	i
10	A	[LINE FEED]	42	2A	*	74	4A	J	106	6A	j
11	B	[VERTICAL TAB]	43	2B	+	75	4B	K	107	6B	k
12	C	[FORM FEED]	44	2C	,	76	4C	L	108	6C	l
13	D	[CARRIAGE RETURN]	45	2D	-	77	4D	M	109	6D	m
14	E	[SHIFT OUT]	46	2E	.	78	4E	N	110	6E	n
15	F	[SHIFT IN]	47	2F	/	79	4F	O	111	6F	o
16	10	[DATA LINK ESCAPE]	48	30	0	80	50	P	112	70	p
17	11	[DEVICE CONTROL 1]	49	31	1	81	51	Q	113	71	q
18	12	[DEVICE CONTROL 2]	50	32	2	82	52	R	114	72	r
19	13	[DEVICE CONTROL 3]	51	33	3	83	53	S	115	73	s
20	14	[DEVICE CONTROL 4]	52	34	4	84	54	T	116	74	t
21	15	[NEGATIVE ACKNOWLEDGE]	53	35	5	85	55	U	117	75	u
22	16	[SYNCHRONOUS IDLE]	54	36	6	86	56	V	118	76	v
23	17	[END OF TRANS. BLOCK]	55	37	7	87	57	W	119	77	w
24	18	[CANCEL]	56	38	8	88	58	X	120	78	x
25	19	[END OF MEDIUM]	57	39	9	89	59	Y	121	79	y
26	1A	[SUBSTITUTE]	58	3A	:	90	5A	Z	122	7A	z
27	1B	[ESCAPE]	59	3B	;	91	5B	[	123	7B	{
28	1C	[FILE SEPARATOR]	60	3C	<	92	5C	\	124	7C	|
29	1D	[GROUP SEPARATOR]	61	3D	=	93	5D	]	125	7D	}
30	1E	[RECORD SEPARATOR]	62	3E	>	94	5E	^	126	7E	~
31	1F	[UNIT SEPARATOR]	63	3F	?	95	5F	_	127	7F	[DEL]

CPSC 100

Computational Thinking

Data Representation in Action

Instructor: Parsa Rajabi
Department of Computer Science
University of British Columbia

Agenda

- Invitation for Research Study
- Course Admin
  - AI Disclosure Form
  - Course check-in survey
- Learning Goals
- Data Representation *in Action*
- Midterm Grades

Research Study

Study Title: Has the clubhouse been unlocked? Exploring experiences of women and non-binary students in computer science education

Investigators: Dr. Susan Gerofsky (Associate Professor, Faculty of Education), Erica Huang (PhD Candidate, Faculty of Education)

Goal: To better understand the CS education experiences of women and non-binary students in first-year CS classes

What's involved?

✅ **Pre-interview survey** (~10 mins)

✅ **In-person interview on campus – Wed. March 19** (<1 hour)
(with flexibility to make alternative arrangement)

✅ **Optional follow-up interview on Zoom** (< 30 mins)

Contact: Erica Huang (eythuang@student.ubc.ca)

Link to pre-interview survey:

Contact: Erica Huang (eythuang@student.ubc.ca)

Course
Admin

Course Admin

- **This week's Lab:** Project Milestone 2
  - Due Wednesday, March 12, 11:59pm
  - **[New!]** *AI Disclosure Form*
- **PC Quiz 5**
  - Due Sunday, March 16, 11:59pm
- **Final Exam**
  - Tuesday, April 22, 7pm; Location TBA

Project
AI Disclosure Form

CPSC 100 - AI Use Disclosure

Why this survey? This disclosure form aims to *help us understand and reduce any potential risks* related to the use of AI tools in this course.

Please be truthful in answering the survey. *Using AI tools will not negatively affect your marks for the assignment.*

Regardless of how your team member(s) have used AI tools, answer the questions based on *your own use* of AI tools for completing *your part* of the assignment.

What do we mean by AI tools in this survey? AI content generators refer to any tools that create any type of content, including:
- ChatGPT and all other large-scale language models;
- Stable Diffusion and other AI-based Image Generators; and
- Other tools that are capable of generating text, image, codes, or any other content for the course work.

BUT, Grammarly and other grammar checkers are NOT included.

Your 8-digit Student Number

First Name (as appears on Canvas)

Last Name (as appear on Canvas)

Link to Form

Each student submits their own disclosure

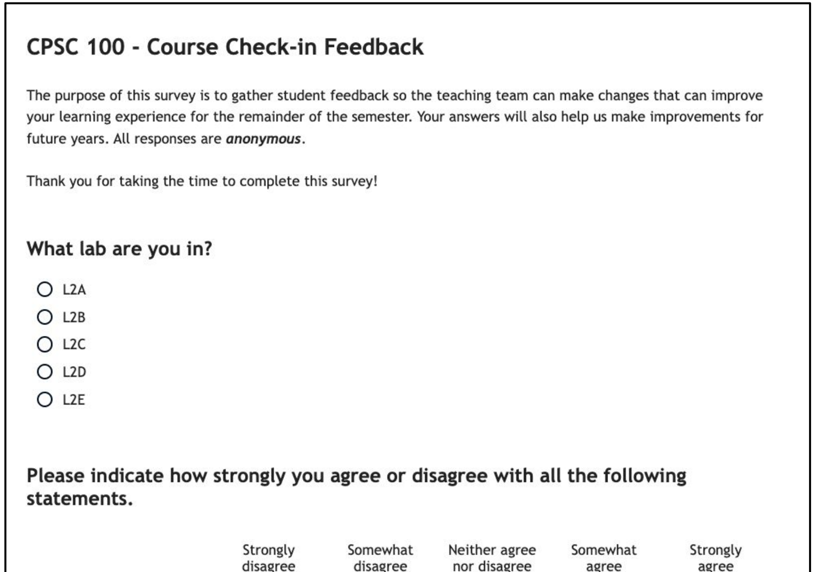
No submission? 10% penalty applied

Also available via Canvas > CPSC 100 > Scroll down to project section

Course Check-in Survey

Course Check-in Survey

- Please fill out the anonymous survey below to provide your thoughts on the course thus far. Your feedback will be used to improve the course!
- https://ubc.ca1.qualtrics.com/jfe/form/SV_26a4t2Ppcw6mJ6u



CPSC 100 - Course Check-in Feedback

The purpose of this survey is to gather student feedback so the teaching team can make changes that can improve your learning experience for the remainder of the semester. Your answers will also help us make improvements for future years. All responses are ***anonymous***.

Thank you for taking the time to complete this survey!

What lab are you in?

- L2A
- L2B
- L2C
- L2D
- L2E

Please indicate how strongly you agree or disagree with all the following statements.

	Strongly disagree	Somewhat disagree	Neither agree nor disagree	Somewhat agree	Strongly agree

Learning Goals

Learning Goals

After this **today's lecture**, you should be able to:

- Understand how colours are represented in the RGB model
- Describe how RGB colours are stored and represented in computing.
  - Recognize that each colour (R, G, B) has 256 intensity levels (0-255).
- Convert between different RGB representations: Decimal, Binary, & Hex

After watching the **take-home video**, you should be able to:

- Analyze how RGB colours mix to produce new colours.
- Explain and apply colour theory concepts to digital designs

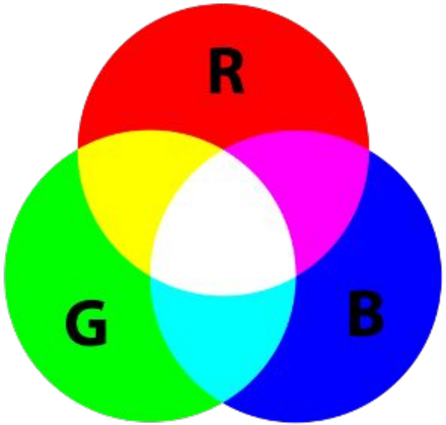
RGB Colours

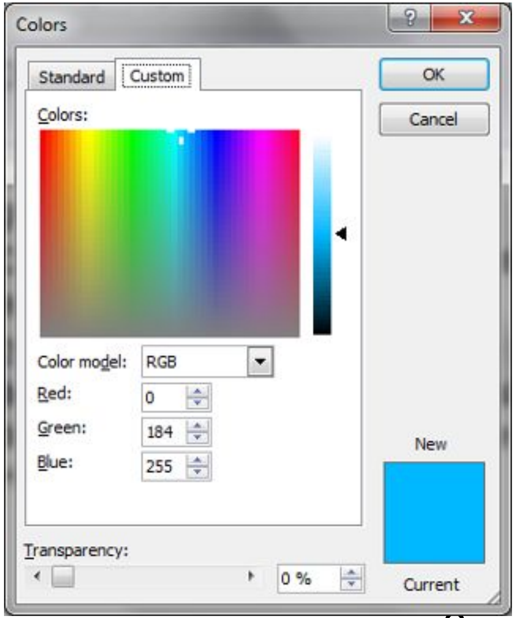
RGB Colours

RGB Colours

- Monitors, phone screens, and TVs make different colours by mixing Red, Green, and Blue lights

- Computer applications use 256 intensities (8 bits) for each of red, green, and blue.





RGB Colours

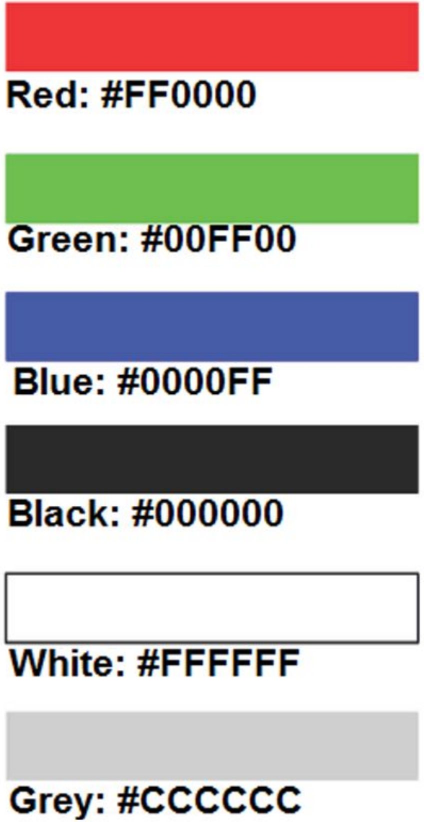
Black is the absence of light:

- 0000 0000 0000 0000 0000 0000 **(Binary)**
- 0 0 0 0 0 0 **(Hex)**
  - RGB bit assignment for black

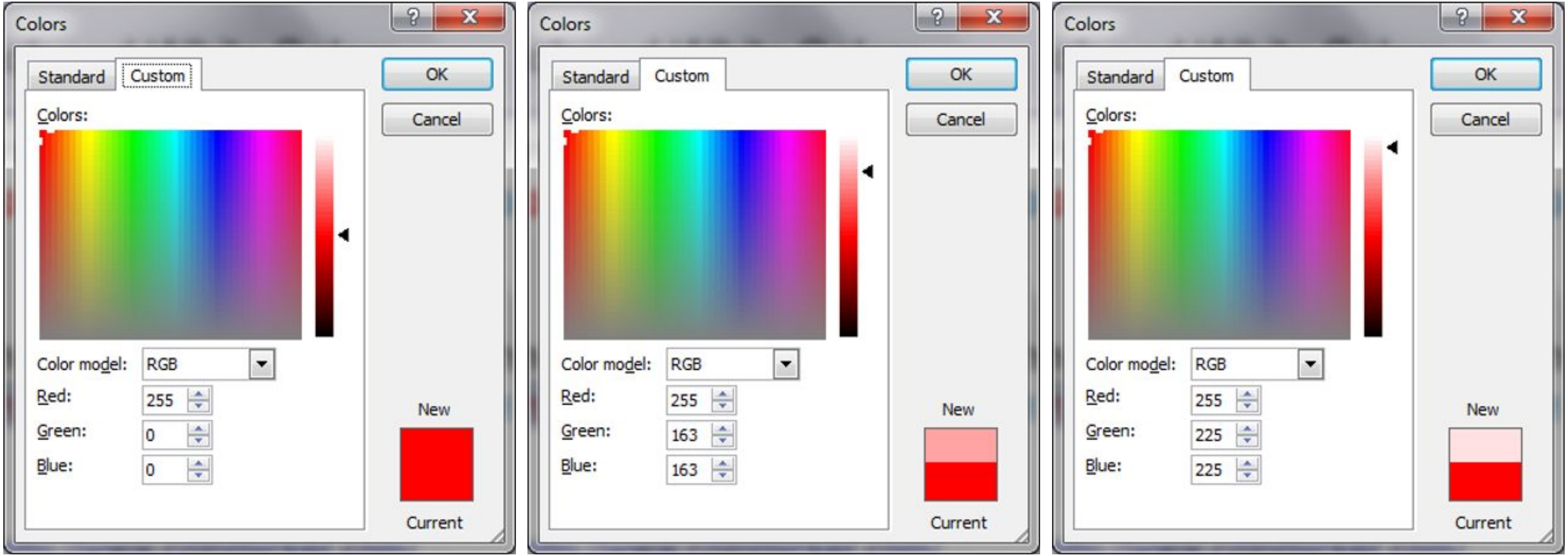
White is the full intensity of each color:

- 1111 1111 1111 1111 1111 1111 **(Binary)**
- F F F F F F **(Hex)**
  - RGB bit assignment for white

- https://www.w3schools.com/colors/colors_picker



RGB Colours in Decimal



RGB Colours

Colour		Decimal		Hex
Red	→	(255, 0, 0)	→	#FF0000
Green	→	(0, 255, 0)	→	#00FF00
Blue	→	(0, 0, 255)	→	#0000FF
White	→	(255, 255, 255)	→	#FFFFFF
Black	→	(0, 0, 0)	→	#000000

RGB Colours

- **Colour** **Decimal** **Hex**
- Red → (255, 0, 0) → #FF0000
- Green → (0, 255, 0) → #00FF00
- Blue → (0, 0, 255) → #0000FF
- White → (255, 255, 255) → #FFFFFF
- Black → (0, 0, 0) → #000000

Recall: Hexadecimal is a base-16 system where:

- *Digits 0-9 represent values 0-9.*
- *Letters A-F represent values 10-15.*

- *Example:* $FF\ (\text{Hex}) = (F \times 16^1) + (F \times 16^0)$

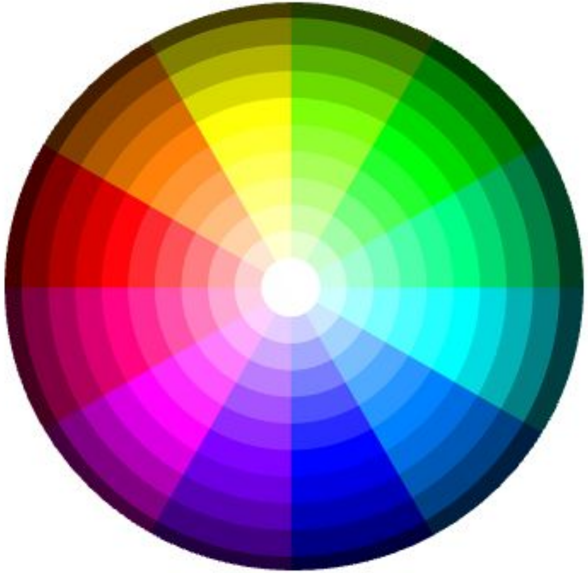
$$(15 \times 16) + (15 \times 1) \Rightarrow 255$$

Decimal	Hex
0	0
1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	A
11	B
12	C
13	D
14	E
15	F

Q: Which colour best describes the one represented by the hexadecimal colour code: #32CD32?

iClicker

A. Shade of red
B. Shade of blue
C. Shade of green
D. Shade of purple
E. Shade of yellow



iClicker

Q: Which colour best describes the one represented by the hexadecimal colour code: #32CD32?

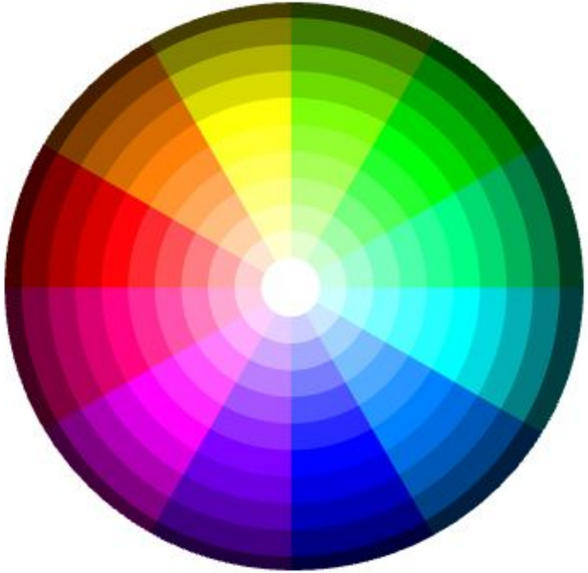
A. Shade of red

B. Shade of blue

C. Shade of green

D. Shade of purple

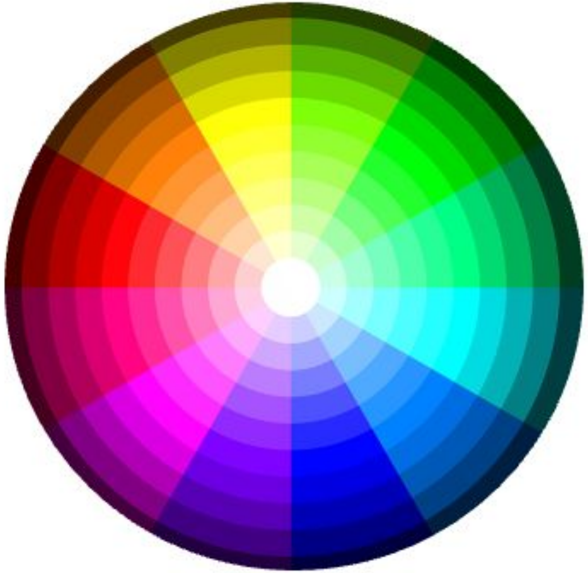
E. Shade of yellow



Q: Which colour best describes the one represented by the hexadecimal colour code: #800B80?

iClicker

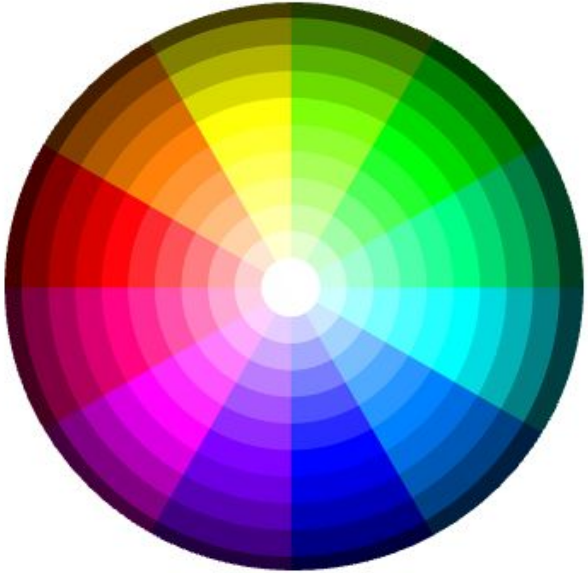
A. Shade of red
B. Shade of blue
C. Shade of green
D. Shade of purple
E. Shade of yellow



iClicker

Q: Which colour best describes the one represented by the hexadecimal colour code: #800B80?

A. Shade of red
B. Shade of blue
C. Shade of green
D. Shade of purple
E. Shade of yellow



Mini-Activity

Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.		shade of ...
#FFA933		Yellow
#FF99FF		Pink or Magenta
#EAE51D		Blue
#A1A2A3		Orange
#1234F8		Grey

Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of ...
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey

Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of ...
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey

Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of …
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey

Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of …
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey

Color Mixing: Match the Colour to its Hex Rep.

Hex. Rep.	shade of ...
#FFA933	Yellow
#FF99FF	Pink or Magenta
#EAE51D	Blue
#A1A2A3	Orange
#1234F8	Grey

Take Home Video

Colour Theory

https://youtu.be/_2LLXnUdUlc?si=ZC0gCVCkhlmnc3KT

What was your main takeaway from today's session?

Do not edit
How to change the design

Presenting with animations, GIFs or speaker notes? Enable our Chrome extension

slido

Wrap up

Wrap Up

- **This week's Lab:** Project Milestone 2
  - Due Wednesday, March 12, 11:59pm
  - **[New!]** *AI Disclosure Form*
- **PC Quiz 5**
  - Due Sunday, March 16, 11:59pm
- **Final Exam**
  - Tuesday, April 22, 7pm; Location TBA